

Project Title

Mutation Recurrence and Evolutionary Constraint in Colorectal
Cancer Driver Genes: A Residue-Level Analysis

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Abstract

Mutation retention and clonal expansion within tumors are shaped by positive selection, not random processes. Over evolutionary time, purifying selection conserves functionally critical residues across species. If recurrent cancer mutations preferentially occur at these constrained sites, this would link macroevolutionary constraint with microevolutionary somatic selection, distinguishing strongly selected mutations from background variation within the same genes. Somatic mutation data from 526 colorectal adenocarcinoma tumors (TCGA COADREAD) were analyzed across APC, TP53, KRAS, BRAF, SMAD4, TGFBR2, PTEN, and PIK3CA. From 1,579 mutation events, 263 unique residue positions were retained after restricting to missense substitutions and collapsing by position; 15 were recurrent (≥ 5 tumors) and 248 non-recurrent. Per-residue conservation scores were computed from multi-species vertebrate alignments via Clustal Omega. Mann-Whitney U testing found no significant difference between groups ($p = 0.24$), reflecting limited power given only 15 recurrent residues and compressed conservation scores. Spearman correlation revealed a weak but significant positive association between recurrence magnitude and conservation score ($\rho = 0.139$, $p = 0.025$), indicating residues mutated in more tumors tend to occur at more constrained positions. Logistic regression confirmed the direction of this association (coefficient = 6.2, $p = 0.27$, Pseudo $R^2 = 0.025$). Oncogenes KRAS, BRAF, and PIK3CA exhibited concentrated hotspot patterns consistent with convergent somatic evolution: KRAS G12D was observed in 57 independent tumors, BRAF V600E in 48, and PIK3CA E545K in 35, reflecting gain-of-function selection at specific activating residues. Tumor suppressors APC and TGFBR2 showed entirely dispersed profiles with no recurrent residues, consistent with loss-of-function mechanisms tolerating broad mutational diversity. These findings suggest residue-level recurrence and conservation data may support evidence-based gene prioritisation in colorectal cancer molecular diagnostic panels.

1. Introduction

1.1 Mutation, Drivers, and Somatic Selection

Carcinogenesis arises from genomic instability within somatic cellular lineages, encompassing alterations at multiple biological levels including nucleotide sequence, chromatin regulation, and cellular state. These processes result in an elevated mutation rate and the progressive accumulation of genetic alterations, generating substantial heterogeneity across malignancies (Brown et al., 2019; McFarland et al., 2017). Among these, somatic mutations at the nucleotide sequence level represent the most directly interpretable alterations in terms of recurrence, selective pressure, and functional consequence at the residue level. Although the vast majority of such mutations are functionally neutral and classified as passenger mutations, a small subset confers a proliferative or survival advantage to the affected cell. These driver mutations promote clonal expansion and contribute directly to tumor initiation, progression, invasion, and metastasis (Brown et al., 2019; Fontana et al., 2023). Cancer genomics has demonstrated that most tumors harbor only a limited number of driver mutations, often two to eight, amid thousands of passenger alterations, making the identification of true driver events a central challenge in cancer research (Chillón-Pino et al., 2024). Because mutation processes are largely stochastic, each tumor represents an independent evolutionary experiment in which mutations arise randomly (Wong et al., 2022), while somatic selection acts on mutations that enhance cellular fitness. In this framework, mutation generation is largely blind to protein function, whereas mutation retention and clonal expansion are non-random and shaped by positive selection acting within the tumor (Fontana et al., 2023; Wong et al., 2022).

1.2 Recurrence, Constraint, and Functional Interpretation

The distribution of mutations within cancer genomes is highly non-uniform. Certain genes, including oncogenes and tumor suppressors, are recurrently mutated across patients, and specific amino acid positions, referred to as mutation hotspots, are disproportionately affected (Hoyos et al., 2022). The repeated occurrence of identical residue-level mutations in independent tumors suggests positive selection acting on functionally advantageous alterations. However, recurrence alone is not definitive evidence of functional importance, as some genomic sites may be preferentially mutated due to intrinsic mutational biases (Bányai et al., 2021; Hoyos et al., 2022; Persi et al., 2025; Wong et al., 2022). Distinguishing selection-driven recurrence from mutation-driven recurrence therefore remains a critical challenge in cancer genomics (Tang et al., 2021). One biologically grounded approach to addressing this distinction lies in the concept of evolutionary constraint, which can serve as a proxy for functional importance (Wollenberg Valero, 2020). Proteins are not merely linear amino acid sequences but structured molecules composed of domains, active sites, binding interfaces, and regulatory regions that are structurally and functionally intolerant to substitution. Mutations affecting these critical regions are more likely to disrupt protein function (Chen et al., 2025; Wollenberg Valero, 2020).

Over evolutionary time, purifying selection removes deleterious changes at functionally important residues, resulting in conserved amino acids across species. This evolutionary conservation therefore reflects underlying structural and functional constraints within proteins (Murphy et al., 2022; Zhao et al., 2019). Because protein function is primarily encoded at the level of individual amino acid residues, mutation-level analyses provide a higher resolution view of selective pressures within driver genes than gene-level approaches alone. If recurrent cancer mutations preferentially occur at evolutionarily conserved and structurally constrained residues, this would support the interpretation that somatic selection systematically exploits functionally critical sites rather than reflecting purely mutational processes (Brown et al., 2019; Greaves, 2015). Integrating evolutionary conservation as a proxy for structural and functional constraint therefore provides a principled framework for interpreting mutation recurrence within cancer driver genes (Chang et al., 2016).

1.3 Mutation-Level Analysis Within Driver Genes

In cancer genomics, driver genes are commonly identified through mutation enrichment relative to estimated background mutation rates. A gene is considered enriched when it is mutated more frequently than expected based on its baseline mutation rate (Chillón-Pino et al., 2024; Supek et al., 2014; Tang et al., 2021). Modeling background mutation processes is therefore essential for distinguishing significantly mutated genes, sub-gene regions, and mutational hotspots from stochastic variation. However, this modeling is complicated by the fact that mutation frequency varies substantially across genes due to factors such as gene length, chromatin accessibility, replication timing, and local mutational exposure, characteristics that are unrelated to functional selection. These gene-level differences introduce confounding factors that complicate interpretation when comparing mutation frequencies across genes (Brown et al., 2019). Moreover, most large-scale cancer genomics studies focus on identifying candidate driver genes through between-gene comparisons, and systematic evaluation of selective patterns within individual genes remains more limited (Chang et al., 2016; Nourbakhsh et al., 2024). Analyzing recurrent and non-recurrent mutations within the same gene, rather than across genes, holds these confounding characteristics constant and isolates variation at the residue level where somatic selection most directly operates (Nourbakhsh et al., 2024).

1.4 Colorectal Cancer as a Model System

Colorectal cancer was selected as a model system due to its substantial global disease burden and well-characterized mutational landscape. Comprehensive genomic profiling efforts have identified recurrent alterations in key oncogenes and tumor suppressor genes that define colorectal tumorigenesis, providing a biologically grounded context in which to investigate mutation-level selection. Publicly available resources such as COSMIC and cBioPortal offered extensive mutation-level data from large patient cohorts, enabling rigorous and reproducible computational analysis (Cerami et al., 2012; Gao et al., 2013; Klimeck et al., 2023). This study obtained the somatic mutation data from the Colorectal Adenocarcinoma cohort of The Cancer Genome Atlas Pan-Cancer Atlas (TCGA), accessed via cBioPortal (study ID: coadread_tcga_pan_can_atlas_2018) (The Cancer Genome Atlas Research Network, 2018), and

focused on a curated panel of eight well-established colorectal cancer driver genes (APC, TP53, KRAS, BRAF, SMAD4, TGFBR2, PTEN, and PIK3CA), allowing detailed examination of residue-level recurrence patterns within a defined biological framework.

1.5 Study Aim

The overall aim of this project is to test whether mutation recurrence within established colorectal cancer driver genes reflects selective disruption of evolutionarily conserved protein regions, rather than background mutational processes. The objective is not to assume that recurrent mutations are important solely because they recur, but to determine whether evolutionary conservation provides a measurable statistical signal that distinguishes strongly selected mutations from background variation within the same genes. If supported, this framework would conceptually link macroevolutionary constraint across species with microevolutionary somatic selection within tumors, moving the analytical focus from identifying driver genes to understanding which specific mutations within those genes are under selection and why. Practically, such an approach may improve the interpretation of rare or emerging variants, support conservation-informed clinical annotation, and help distinguish between mutations that are merely frequent and those that disrupt deeply conserved functional regions.

2. Methods

To clarify the hierarchical structure of the mutation dataset and the analytical steps applied throughout the study, Table 1 summarizes the data processing workflow, the unit of analysis at each stage, and the corresponding results presented in subsequent sections.

Table 1*Data Processing Workflow and Analytical Units Applied Across the Study*

Step	Dataset level	Description	Unit of analysis	N	Analyses using this level
1	Tumors	Independent colorectal cancer samples from the TCGA Pan-Cancer Atlas cohort	Tumor	526	Cohort description
2	Tumor-gene mutation occurrences	Presence of at least one mutation in a driver gene within a tumor; wild-type entries removed	Tumor-gene pair	1,243	Mutation burden per gene (Table 2)
3	Mutation events	Individual protein-altering mutations identified across tumors; multi-mutation cells split	Mutation event	1,579	Recurrence histogram (Figure 1); recurrence classification (Table 4); mutation subtype distribution (Table 5)
4	Missense mutation events	Subset of mutation events corresponding to single amino acid substitutions only	Mutation event	846	Missense distribution (Table 6)
5	Unique missense mutation identities	Distinct amino acid substitutions; duplicates removed. Each substitution represented once with recurrence count attached	Mutation identity	724	Recurrence distribution analysis; most recurrent mutations (Table 3)

6	Unique mutated residue positions	Missense mutations collapsed by gene-position pair; conservation scores merged from multi-species alignment	Residue position	263	Spearman correlation; Mann-Whitney test; logistic regression; conservation statistics (Table 7); recurrence vs conservation scatter plot (Figure 2)
7	Recurrent residues	Residues mutated in five or more independent tumors	Residue position	15	Mann-Whitney test; logistic regression; conservation comparison (Table 7)
8	Gene-level mutation architecture	Distribution of mutations along protein sequences per gene	Residue position	—	Recurrence vs residue position plots (Figure 4)
9	Recurrence architecture per gene	Distribution of recurrence categories across genes	Residue position	—	Recurrence heatmap (Figure 3)
10	Hotspot concentration	Fraction of recurrent residue positions per gene	Gene	8 genes	Recurrent fraction barplot (Figure 5)
11	Mutation dispersion across proteins	Number of unique mutated residue positions per gene	Gene	8 genes	Mutated residues per gene barplot (Figure 6)

Note. All processing performed in Python (pandas, scipy, statsmodels, seaborn) within Google Colab. Multi-species sequence alignments generated using Clustal Omega via EMBL-EBI. Steps 1–7 represent progressive refinements of the analytical dataset. Steps 8–11 represent derived summary analyses conducted at the residue and gene levels.

2.1 Data Source and Cohort Selection

Somatic mutation data were obtained from the Colorectal Adenocarcinoma cohort of The Cancer Genome Atlas (TCGA) Pan-Cancer Atlas, accessed through cBioPortal (study ID: coadread_tcga_pan_can_atlas_2018; Cerami et al., 2012; Gao et al., 2013). The full cohort comprised 594 colorectal tumor samples; however, after filtering to retain only samples harboring at least one mutation in the eight selected driver genes, 526 samples were included in the final analysis, constituting the primary cohort analyzed in this study (Table 1, Step 1). Eight established driver genes with well-documented roles in colorectal tumorigenesis were selected: APC, TP53, KRAS, BRAF, SMAD4, TGFBR2, PTEN, and PIK3CA. Mutation data were downloaded as the file mutations_transposed.txt from the raw mutation section of cBioPortal. In the original file, genes were arranged as rows and tumor samples as columns.

2.2 Data Reshaping and Cleaning

All data processing and statistical analyses were performed in Python using the packages pandas, NumPy, SciPy, statsmodels, matplotlib, seaborn, and re within the Google Colab environment. The SAMPLE_ID row was promoted to column headers, converting sample identifiers into column names. To facilitate downstream analyses, the wide-format dataset was reshaped into a long format using the pandas.melt() function. Wild-type entries were removed, retaining only mutated observations. At this stage, each row represented a single tumor in which a specific gene contained at least one mutation, regardless of the number of mutations present within that gene. This constituted the first analytical level of the study and resulted in 1,243 tumor–gene mutation occurrences (Table 1, Step 2). In other words, a single tumor such as one carrying mutations in both APC and KRAS would contribute two separate tumor-gene occurrences to this count.

2.3 Mutation Event Construction

The second unit of analysis was generated by separating entries containing multiple mutations within the same gene and tumor into individual records using str.split() followed by pandas.explode(). This procedure converted each protein-altering mutation into its own row,

yielding 1,579 mutation events (Table 1, Step 3), with each row representing a single mutation occurring in a specific tumor sample. The increase from 1,243 tumor–gene mutation occurrences to 1,579 mutation events reflects the presence of multiple mutations within the same gene in some tumors. For example, a tumor harboring both APC R1450* and APC E1309D would initially be represented by a single tumor–gene mutation occurrence but would subsequently be expanded into two separate mutation-event records, one for each protein-altering mutation.

Residue positions were then extracted from the protein-change annotations by parsing the amino acid substitution notation, such that G12D was mapped to residue position 12. Mutations were subsequently classified into four categories: missense, nonsense, frameshift, and other, using a rule-based parsing approach applied to the protein-change annotations. This classification enabled downstream analyses of mutation subtype distributions, residue-level recurrence, and evolutionary conservation.

2.4 Recurrence Quantification and Classification

Recurrence was quantified at the level of individual mutation identities. A mutation identifier (mutation_id) was generated by concatenating the gene name and protein change annotation (e.g., KRAS_G12D). For each unique mutation identifier, recurrence was defined as the number of independent tumor samples harboring that specific mutation. Recurrence values were calculated by grouping observations according to mutation_id and counting the number of unique sample identifiers associated with each mutation using the nunique() function applied to the Sample_ID column. The resulting recurrence values were merged back into the complete mutation-event dataset (N = 1,579), ensuring that each mutation event carried the recurrence count corresponding to its mutation identity. Prior to classification, the distribution of recurrence values was visualized using a histogram (Figure 1) to characterize the overall recurrence landscape.

A recurrence threshold of five or more independent tumors (≥ 5) was then applied to classify mutations as recurrent or non-recurrent. This threshold was selected to distinguish well-supported hotspot mutations from low-frequency events more likely to represent background variation, while retaining sufficient observations in both groups for downstream statistical analyses.

2.5 Restriction to Missense Mutations

All conservation analyses were restricted to missense mutations (Table 1, Step 4; N = 846 mutation events), as only single amino acid substitutions can be directly mapped to specific residue positions for evolutionary conservation analysis. Nonsense and frameshift mutations were retained for descriptive characterization of mutation subtypes (Table 5) but were excluded from conservation analyses because they alter protein function through premature truncation or reading-frame shifts rather than through substitution of a specific amino acid residue.

2.6 Generation of Unique Mutation and Residue-Level Datasets

To prepare the dataset for conservation analysis, the missense mutation table underwent two successive collapsing steps (Table 1, Steps 5 and 6).

First, duplicate mutation entries were removed by applying `drop_duplicates()` to the `mutation_id` and `recurrence` columns, reducing the dataset from 846 missense mutation events to 724 unique mutation identities (Table 1, Step 5). At this stage, each distinct amino acid substitution appeared only once, with the number of independent tumors harboring that mutation recorded as its recurrence value. For example, KRAS G12D, which appeared in 57 independent tumors, was represented as a single row with a recurrence value of 57. Of these 724 unique mutation identities, 47 recurred in five or more tumors, whereas 677 occurred in fewer than five tumors.

Second, the dataset was collapsed to unique residue positions. Because evolutionary conservation is a property of a residue position rather than a specific amino acid substitution, multiple mutations affecting the same position within a gene were combined into a single record. For example, mutations occurring at codon 12 of KRAS, such as G12D (57 recurrences), G12V (49 recurrences), and G12C (15 recurrences), were merged into a single residue-level entry by grouping on gene name and residue position. When multiple substitutions occurred at the same position, the highest recurrence value observed among those substitutions was retained. In this example, codon 12 of

KRAS retained a recurrence value of 57, corresponding to KRAS G12D and representing the strongest observed recurrence signal at that site.

This second collapsing step reduced the dataset from 724 unique mutation identities to 263 unique residue positions (Table 1, Step 6). The resulting residue-level dataset constituted the final analytical dataset used for all conservation analyses and statistical comparisons.

2.7 Reference Sequence Retrieval

The canonical human protein sequence for each of the eight selected driver genes was retrieved in FASTA format from the UniProt database using the UniProt REST API. Queries were performed using the gene name and restricted to Homo sapiens (taxonomy ID: 9606), with the reviewed filter applied to return only manually curated Swiss-Prot entries. The accession numbers of the retrieved reference sequences were APC (P25054), TP53 (P04637), KRAS (P01116), BRAF (P15056), SMAD4 (Q13485), TGFBR2 (P37173), PTEN (P60484), and PIK3CA (P42336). Prior to alignment, the ungapped length of each retrieved human sequence was verified against the canonical expected length, confirming that the correct full-length isoform had been retrieved and that mutation coordinates from cBioPortal would map correctly onto the reference sequence.

2.8 Ortholog Retrieval and Filtering

Vertebrate orthologous protein sequences were retrieved for each gene using the UniProt REST API by querying the gene name against the vertebrate taxonomy (taxonomy ID: 7742), with a maximum of 800 entries returned per query. Searches were initially restricted to reviewed Swiss-Prot entries. When insufficient ortholog coverage was available, the filter was relaxed to include unreviewed TrEMBL entries. Preference was given to reviewed sequences whenever possible, as these entries are manually curated and are less likely to contain annotation errors, incomplete sequences, or redundant isoforms.

Three sequential filtering steps were then applied. First, a length filter of $\pm 30\%$ relative to the human reference protein was used to exclude likely fragments and highly divergent isoforms. Sequences with substantial length differences can introduce excessive alignment gaps and negatively affect residue-level conservation estimates. Second, only one sequence per species was retained. When multiple entries were available for the same species, the sequence exhibiting the highest identity to the human reference protein was selected. This step prevented individual species from contributing disproportionately to the conservation calculation. Third, species selection prioritized phylogenetic diversity. For each gene, the final sequence set preferentially included at least one bird (Aves, taxonomy ID: 8782) and one ray-finned fish (Actinopterygii, taxonomy ID: 7898) whenever suitable orthologs were available. The remaining sequences were selected from mammals and, where possible, amphibians. This strategy ensured representation across a broad evolutionary range and reduced the risk of conservation estimates being biased by an overrepresentation of closely related species. The final alignment dataset comprised eight vertebrate sequences per gene, including the human reference sequence.

2.9 Multiple Sequence Alignment and Conservation Scoring

For each of the eight genes, the combined FASTA file containing the human reference sequence and up to seven vertebrate orthologs was submitted to the Clustal Omega multiple sequence alignment tool (EMBL-EBI) using default parameters and FASTA alignment output. The same alignment procedure was applied uniformly across all genes.

Conservation scores were subsequently derived from each alignment using a column-scanning approach implemented in Python. For each alignment, the human sequence was first scanned to construct a mapping between human residue positions and their corresponding alignment columns. This was achieved by traversing the aligned human sequence character by character, incrementing the human residue position counter only when the character was not a gap and recording the alignment column index for each non-gap position. This mapping ensured that residue positions derived from cBioPortal mutation annotations, such as position 12 in KRAS G12D, corresponded to the correct alignment column even in the presence of gaps introduced during the alignment process.

For each alignment column corresponding to a human residue position, the conservation score was calculated as the proportion of aligned sequences carrying the same amino acid as the human reference at that position, producing a score between 0 and 1, where a value of 1 indicates complete amino acid identity across all compared species. This procedure generated a per-gene table containing human residue positions and their corresponding conservation scores.

The resulting conservation table contained the gene name, human residue position, and conservation score for each aligned residue position. Conservation scores were subsequently merged with the mutated residue position dataset using gene name and residue position as matching keys, assigning a conservation score to each of the 263 unique mutated residue positions. This produced the final analytical dataset, referred to throughout as `residue_tbl`, comprising the gene name, residue position, recurrence count, conservation score, and recurrence classification for each position. This table served as the input for all subsequent statistical analyses.

2.10 Statistical Analyses

All statistical analyses were conducted at the unique mutated residue position level using `residue_tbl` as input ($N = 263$). A binary classification column was added to the table, assigning a value of 1 to residues with recurrence of five or more tumors and 0 to all remaining residues, consistent with the threshold defined in section 2.5. This produced the two analytical groups used in the distributional comparison: 15 recurrent residues and 248 non-recurrent residues (Table 1, Step 7).

Three statistical tests were performed. A Mann-Whitney U test (two-sided; `scipy.stats.mannwhitneyu`) compared conservation score distributions between the recurrent and non-recurrent groups. This non-parametric test was selected because conservation scores are bounded between 0 and 1 and right-skewed toward high values, and the severe imbalance between groups renders the normality assumption required by a Student's t-test untenable.

A Spearman rank correlation (`scipy.stats.spearmanr`) evaluated the monotonic association between recurrence count and conservation score across all 263 positions. Unlike the binary classification used in the Mann–Whitney analysis, this approach retained the full recurrence count for each residue and therefore assessed whether conservation tended to increase with increasing mutation recurrence.

Finally, a binary logistic regression (`statsmodels.api.Logit`) modelled the probability of a residue being recurrent as a function of its conservation score, with the recurrent binary variable as the outcome. Model coefficients, p-values, and McFadden's pseudo- R^2 values were reported to evaluate the strength and explanatory power of the association between conservation and recurrence.

3. Results

3.1 Mutation Landscape of Colorectal Cancer Driver Genes

The study analyzed 526 tumors from the Pan-Cancer Atlas (TCGA) Colorectal Adenocarcinoma cohort. Eight established driver genes were investigated: APC, TP53, KRAS, BRAF, SMAD4, TGFBR2, PTEN, and PIK3CA. Across these genes, a total of 1,243 tumor–gene mutation occurrences were identified (Table 2).

Table 2

Mutation Occurrences per Gene Across 526 Colorectal Tumor Samples (TCGA COADREAD Cohort)

Gene	Gene class	Tumor-gene mutation occurrences (n)
APC	Tumor suppressor	385
TP53	Tumor suppressor	313
KRAS	Oncogene	217
PIK3CA	Oncogene	146
SMAD4	Tumor suppressor	68
BRAF	Oncogene	62
PTEN	Tumor suppressor	34
TGFBR2	Tumor suppressor	18
Total	—	1,243

Among the selected genes, APC exhibited the highest gene-level mutation burden, being mutated in 385 of the 526 tumors, followed by TP53 (313 tumors) and KRAS (217 tumors).

After resolving multi-mutation entries into individual protein-altering events, the dataset expanded to 1,579 mutation events. Table 3 presents the most recurrent residue-level mutation per gene.

Table 3*Most Recurrent Residue-Level Mutations Across the Selected Driver Genes*

Gene	Gene class	Top mutation	Residue position	Recurrence (n tumors)
KRAS	Oncogene	G12D	12	57
BRAF	Oncogene	V600E	600	48
PIK3CA	Oncogene	E545K	545	35
APC	Tumor suppressor	R1450*	1450	34
TP53	Tumor suppressor	R175H	175	32
SMAD4	Tumor suppressor	R361H*	361	11
PTEN	Tumor suppressor	R130Q*	130	5
TGFBR2	Tumor suppressor	K153N*	153	2

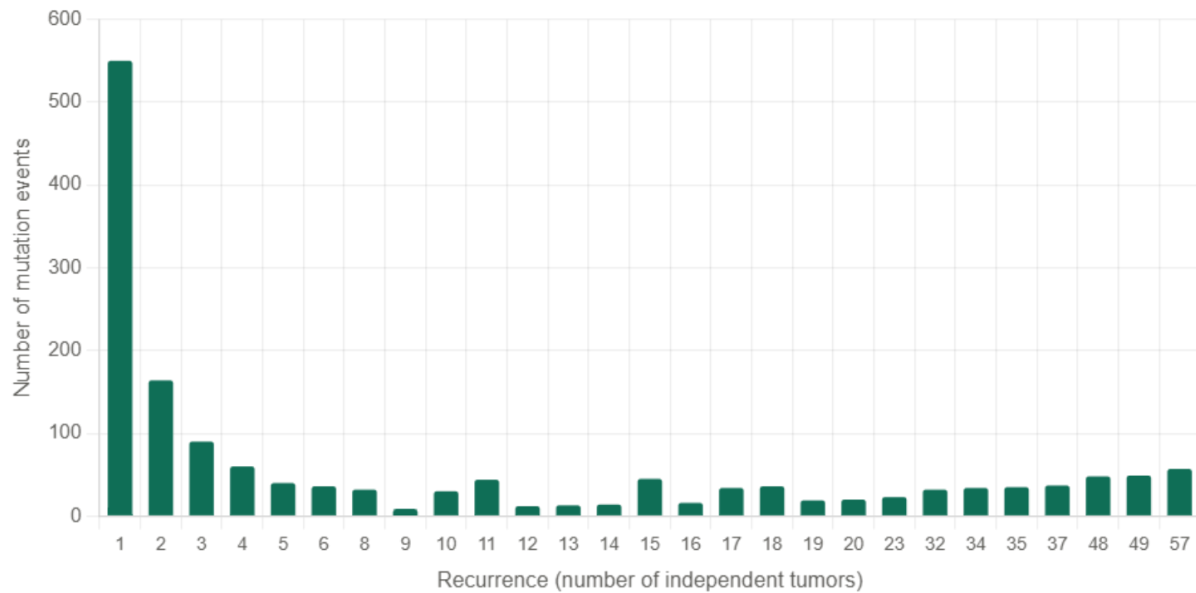
Note. Nonsense mutations retained for comparative reference. Recurrence values reflect the number of independent tumors harboring that specific substitution.

Notably, 57 independent colorectal tumors carried the KRAS G12D substitution, consistent with its well-established role as a canonical colorectal cancer hotspot mutation. In contrast to dominant hotspots such as KRAS codon 12, TP53 R273, and BRAF V600E, mutations in TGFBR2 exhibited markedly lower recurrence. No strong residue-level hotspots were observed, with only two amino acid substitutions recurring more than once.

To illustrate the distribution of recurrence across the 1,579 mutation events, a histogram was generated (Figure 1).

Figure 1

Distribution of Mutation Recurrence Across Colorectal Cancer Samples



Note. Each bar represents the number of mutation events observed at a given recurrence value. The x-axis shows recurrence count (number of independent tumors carrying the same mutation); the y-axis shows the number of mutation events at that recurrence level. N = 1,579 mutation events. Maximum recurrence = 57, corresponding to KRAS G12D.

Figure 1 reveals a heavy-tailed distribution of mutation recurrence. The majority of mutations were observed in only a single tumor (recurrence = 1), producing a pronounced spike, with 550 mutation events occurring only once. In contrast, only a small number of mutations were observed repeatedly across multiple tumors, forming a long right-hand tail extending to a maximum recurrence of 57, corresponding to KRAS G12D. This pattern is consistent with the hotspot mutations identified in Table 3. While most mutations occurred at very low frequencies, a relatively small subset displayed substantial recurrence across independent tumors, highlighting a distinct group of candidate hotspot mutations for subsequent analysis.

Table 4 summarises the application of the recurrence threshold to the 1,579 mutation events, separating hotspot mutations from low-frequency events. Of these, 715 events were classified as recurrent and 864 as non-recurrent. Each row in Table 4 represents a single mutation event

occurring in one tumor. For example, the 57 independent occurrences of KRAS G12D each contribute separately to the recurrent event count for KRAS.

Table 4

Distribution of Mutation Events by Recurrence Class Across the Eight Driver Genes

Gene	Gene class	Non-recurrent (n)	Recurrent (n)	Total (n)
APC	Tumor suppressor	404	233	637
TP53	Tumor suppressor	190	141	331
KRAS	Oncogene	30	192	222
PIK3CA	Oncogene	97	80	177
SMAD4	Tumor suppressor	69	11	80
PTEN	Tumor suppressor	38	10	48
BRAF	Oncogene	18	48	66
TGFBR2	Tumor suppressor	18	0	18
Total	—	864	715	1,579

Note. Recurrent: five or more independent tumors carrying the same substitution. Non-recurrent: fewer than five tumors. Genes ordered by total mutation burden.

KRAS and BRAF exhibited a marked predominance of recurrent events, consistent with strong hotspot-driven selection, whereas APC, TP53, PTEN, and SMAD4 exhibited a predominance of non-recurrent events, reflecting more dispersed mutational patterns. Consistent with the residue-level observations presented in Table 3, TGFBR2 did not exhibit any mutations exceeding the five-tumor recurrence threshold.

The 1,579 mutation events were subsequently classified according to mutation subtype, comprising 846 missense, 444 nonsense, 236 frameshift, and 53 other mutation types. These distributions are summarized in Table 5.

Table 5*Distribution of Mutation Subtypes Across the Analyzed Cohort*

Mutation subtype	Count (n)	Proportion (%)
Missense	846	53.6
Nonsense	444	28.1
Frameshift	236	14.9
Other	53	3.4
Total	1,579	100.0

Note. Missense mutations were retained for conservation analysis. Nonsense, frameshift, and other subtypes were retained for descriptive characterization only.

Table 5 reveals substantial heterogeneity in mutation subtype composition. Missense substitutions represented the largest mutation category, accounting for more than half of all mutation events. A considerable proportion of mutations were also truncating alterations, including nonsense and frameshift mutations, highlighting the diverse mechanisms through which the selected driver genes are altered in colorectal cancer.

3.2 Recurrence Patterns of Missense Mutations

Subsequent analyses focused exclusively on the 846 missense mutation events. Application of the recurrence threshold of five or more tumors classified 456 events as recurrent (53.9%) and 390 as non-recurrent (46.1%). The majority of missense mutation events therefore occurred at residues classified as recurrent, constituting the primary dataset for subsequent conservation analyses.

Table 6 summarizes the distribution of missense mutation events across the eight driver genes.

Table 6*Distribution of Missense Mutation Events per Gene Across the 846 Missense Mutation Subset*

Gene	Gene class	Missense events (n)	Proportion (%)
TP53	Tumor suppressor	235	27.8
KRAS	Oncogene	220	26.0
PIK3CA	Oncogene	172	20.3
APC	Tumor suppressor	67	7.9
BRAF	Oncogene	60	7.1
SMAD4	Tumor suppressor	60	7.1
PTEN	Tumor suppressor	20	2.4
TGFBR2	Tumor suppressor	12	1.4
Total	—	846	100.0

Note. APC accounts for only 7.9% of missense events despite contributing the highest overall mutation burden in the cohort (Table 2), reflecting the predominance of truncating alterations in this gene.

Among the 846 missense mutation events, TP53, KRAS, and PIK3CA contributed the highest counts, with 235, 220, and 172 events respectively, collectively accounting for over 74% of all missense mutations in the cohort. In contrast, APC accounted for only 67 missense events despite exhibiting the highest overall mutation burden in the cohort (Table 2). This pattern reflects the predominance of truncating rather than missense alterations in APC, a well-established characteristic of colorectal cancer.

Recurrence was subsequently evaluated at the level of unique missense mutation identities. A total of 724 distinct amino acid substitutions were identified, of which 47 recurred in five or more tumors and 677 occurred in fewer than five tumors. For example, KRAS G12D, observed in 57 independent tumors, is represented only once at this analytical level with a recurrence value of 57. These results indicate that recurrence is concentrated within a small number of mutation identities, while most substitutions occur infrequently across the cohort.

3.3 Evolutionary Conservation of Mutated Residues

Conservation scores were derived from multi-species protein sequence alignments across eight vertebrate species, as described in sections 2.7 to 2.9, and mapped to the missense mutation dataset. This process reduced the dataset from 724 unique missense mutation identities to 263 unique mutated residue positions, each assigned a conservation score based on amino acid identity across the aligned species. Table 7 summarizes the conservation statistics for these residues, stratified into two groups based on mutation recurrence: recurrent residues (≥ 5 tumors) and non-recurrent residues (< 5 tumors).

Table 7

Summary Statistics of Evolutionary Conservation Scores Stratified by Recurrence Class (N = 263 Unique Residue Positions)

Statistic	Non-recurrent (n = 248)	Recurrent (n = 15)
Mean	0.936	0.990
Standard deviation	0.171	0.038
Minimum	0.000	0.857
25th percentile	1.000	1.000
Median (50th percentile)	1.000	1.000
75th percentile	1.000	1.000
Maximum	1.000	1.000

Note. Conservation scores range from 0 to 1, where 1 indicates complete amino acid identity across all aligned vertebrate species. The 25th, 50th, and 75th percentiles equal 1.0 in both groups, reflecting a pronounced ceiling effect consistent with the strong evolutionary constraint of established driver genes. Recurrent: ≥ 5 independent tumors; non-recurrent: fewer than five tumors. SD = standard deviation.

Residues harboring recurrent mutations occurred at slightly more conserved positions compared to non-recurrent residues, although the difference was modest. The mean conservation score for

recurrent residues was 0.99, whereas non-recurrent residues exhibited a mean of 0.94. Notably, the majority of residues across both groups displayed conservation scores close to 1.0.

3.3.1 Mann-Whitney U Test: Comparison of Conservation Between Recurrent and Non-Recurrent Residues

A Mann–Whitney U test was performed to compare conservation score distributions between recurrent and non-recurrent residues. The recurrent group comprised 15 residues and the non-recurrent group 248 residues. Recurrent residues exhibited slightly higher conservation scores on average (Table 7); however, the difference between the two groups was not statistically significant ($p = 0.21$), and the null hypothesis of no difference in conservation score distributions between the two groups could not be rejected.

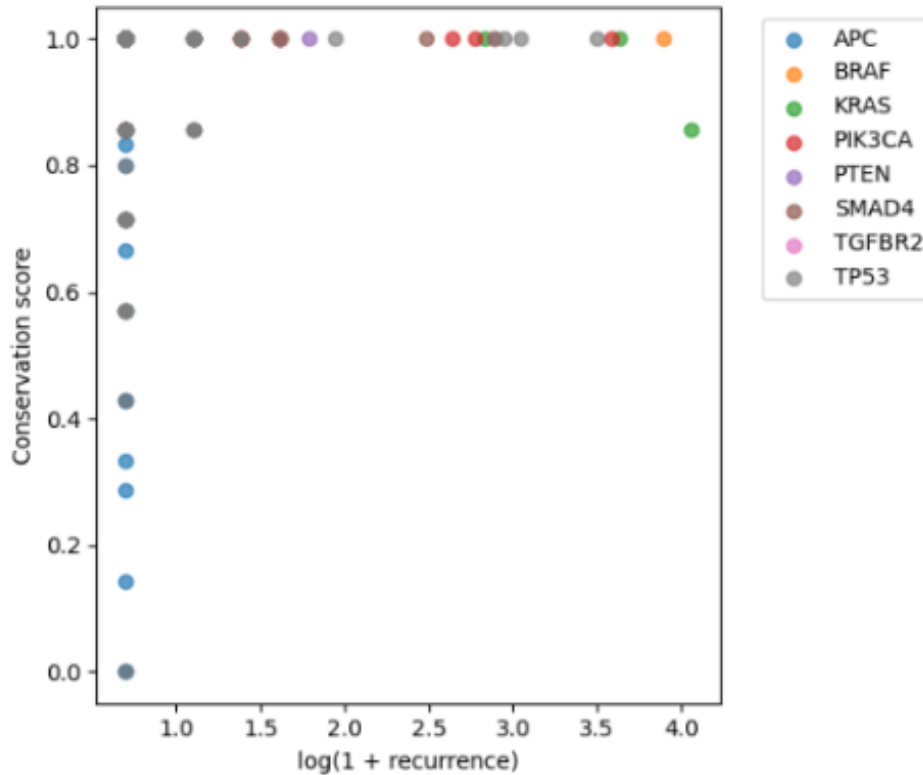
3.4 Spearman Rank Correlation: Association Between Mutation Recurrence and Conservation

To further investigate the relationship between recurrence magnitude and conservation, a Spearman rank correlation was conducted across all 263 mutated residue positions. A weak but statistically significant positive correlation was observed ($\rho = 0.139$, $p = 0.025$), indicating a modest tendency for more frequently mutated residues to occur at more evolutionarily conserved positions. Unlike the Mann–Whitney comparison, which treated recurrence as binary, the Spearman analysis retained the full recurrence magnitude and was therefore able to detect a graded association that the binary classification could not capture.

Figure 2 visually supports the statistical findings by displaying the relationship between mutation recurrence and evolutionary conservation across the 263 unique mutated residue positions. The x-axis represents log-transformed recurrence, calculated as $\log(1 + \text{recurrence})$, to account for the highly skewed distribution of recurrence values, which range from 1 to 57. The y-axis represents the conservation score on a scale from 0 to 1.

Figure 2

Relationship Between Mutation Recurrence and Evolutionary Conservation Score Across 263 Unique Mutated Residue Positions, Colored by Gene



Note. X-axis: $\log(1 + \text{recurrence})$, raw recurrence values 1 to 57. Y-axis: conservation score (0 to 1). Each dot represents one unique residue position. $N = 263$ unique residue positions.

The plot reveals a pattern consistent with the Spearman correlation result. At low recurrence values, conservation scores span the full range from 0.0 to 1.0, reflecting substantial variability among infrequently mutated residues. In contrast, residues with recurrence values above 20, corresponding to log-transformed values greater than approximately 3.0, are exclusively restricted to conservation scores between 0.9 and 1.0. Thus, all highly recurrent mutations occur at highly conserved residue positions, whereas low-recurrence mutations are observed across the full spectrum of conservation scores. Highly recurrent and highly conserved residues were predominantly associated with oncogenes such as KRAS, BRAF, and PIK3CA, whereas tumor suppressor genes, particularly APC and TP53, exhibited a broader spread of low-recurrence mutations across both highly conserved and poorly conserved positions. The gene-specific

coloring further indicates that the observed relationship reflects distinct mutational architectures, with oncogenes clustering toward high recurrence and high conservation and tumor suppressors displaying greater dispersion at lower recurrence levels.

3.4.1 Logistic Regression Analysis

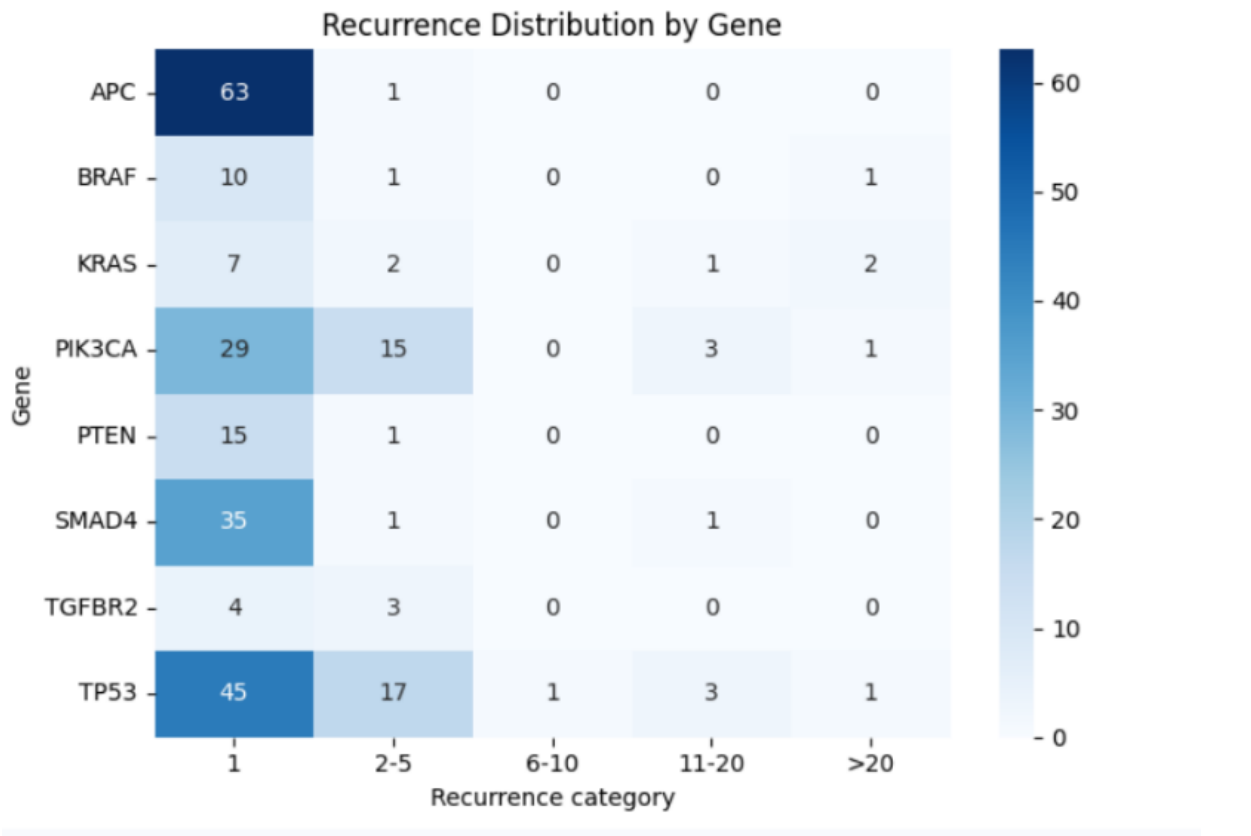
To assess whether conservation score predicts recurrence classification, a binary logistic regression model was fitted with conservation score as the sole predictor. The model yielded a positive coefficient for conservation (coef = 6.215), indicating that higher conservation scores were associated with an increased predicted probability of recurrence. However, the association did not reach statistical significance ($p = 0.274$), and the model produced a McFadden's pseudo R^2 of 0.025, indicating limited explanatory power. These results are consistent with those of the Mann–Whitney test, as the direction of the association was positive across all three analyses but the binary classification approaches lacked sufficient statistical power to detect a significant effect given the small size of the recurrent group and the compressed range of conservation scores across driver gene residues.

3.5 Gene-Specific Mutation Architecture

Heatmap Analysis

To characterize the distribution of recurrence across individual genes, a heatmap was generated displaying the number of mutated residue positions falling within each recurrence category per gene (Figure 3).

Figure 3
Heatmap of Mutated Residue Counts per Recurrence Category Across the Eight Driver Genes (N = 263 Unique Residue Positions)



Note. Each cell displays the number of unique mutated residue positions falling within the indicated recurrence interval. Recurrence category indicates the number of independent tumors carrying the same mutation at a given residue position. Color intensity reflects count magnitude.

APC exhibited a predominantly dispersed mutation pattern, with 63 residues occurring only once, one residue in the 2–5 recurrence category, and no residues exceeding five recurrences. In contrast, KRAS, BRAF, and PIK3CA displayed more concentrated recurrence distributions. Although single-occurrence residues remained the largest category within these genes, they also contained residues distributed across higher recurrence intervals, consistent with the presence of mutational hotspots. KRAS showed the most pronounced hotspot pattern, with two residues exceeding 20 recurrences. PIK3CA exhibited an intermediate architecture, comprising 29 single-occurrence residues, 15 residues in the 2–5 recurrence category, three residues in the 11–20 category, and one residue exceeding 20 recurrences. Together, these patterns illustrate a spectrum ranging from

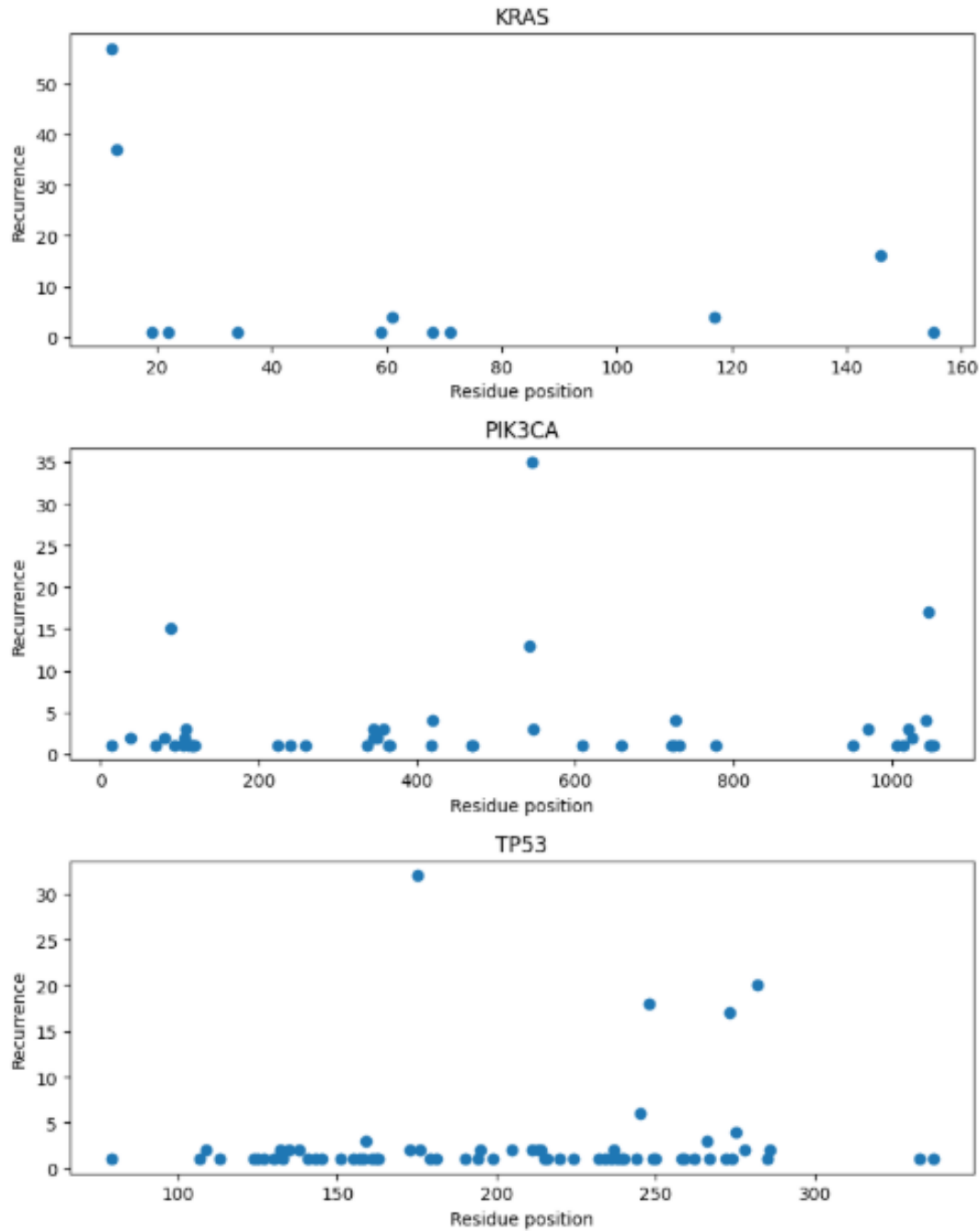
highly dispersed architectures, exemplified by APC, to strongly concentrated hotspot architectures, exemplified by KRAS.

Recurrence Versus Residue Position

To further characterize gene-specific mutation architectures, recurrence count was plotted against residue position for three representative genes (KRAS, PIK3CA, and TP53), selected to illustrate contrasting architectural patterns (Figure 4).

Figure 4

Recurrence Count Plotted Against Residue Position for Three Representative Driver Genes: KRAS, PIK3CA, and TP53



Note. Each dot represents a unique mutated residue position. Y-axis shows the number of independent tumors carrying a mutation at that position. X-axis shows the amino acid residue position within the protein sequence.

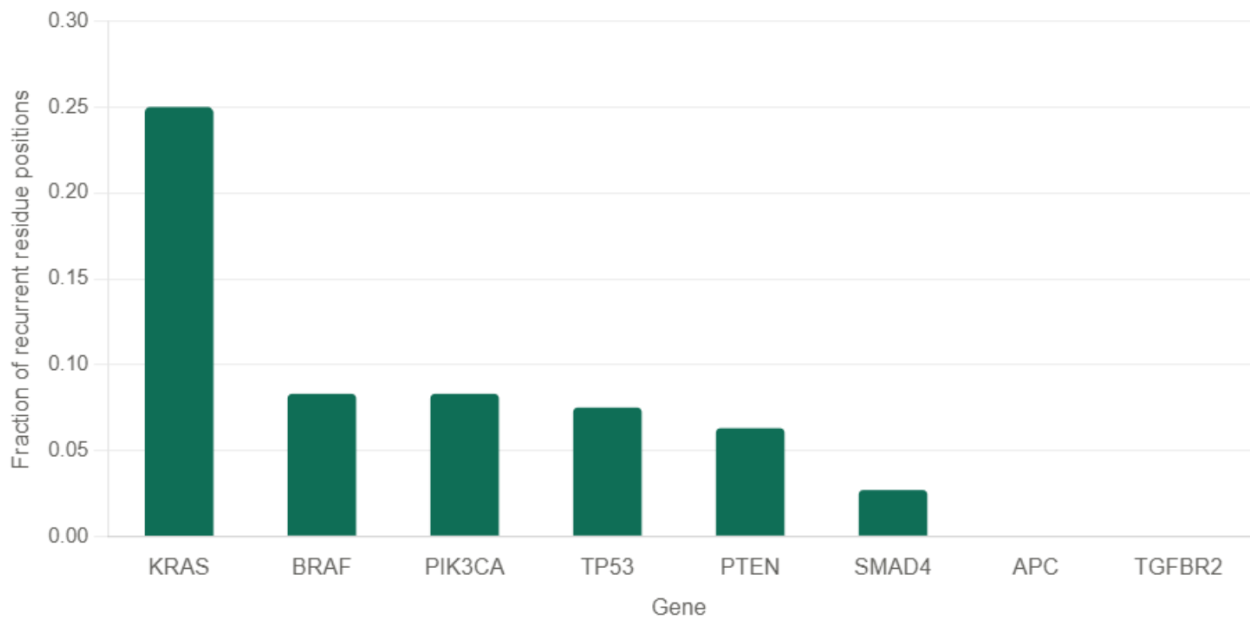
Source: author's own analysis.

Distinct recurrence patterns were observed across the three genes. KRAS exhibited a pronounced peak at codon 12, reaching a recurrence value of 57, consistent with a highly concentrated hotspot pattern. PIK3CA displayed multiple recurrent peaks distributed across different residue positions. TP53 exhibited the broadest distribution of recurrent sites among the three genes, with multiple positions reaching moderate recurrence levels across the DNA-binding domain, consistent with the mixed architectural behavior observed in Figure 3.

Recurrent Fraction per Gene

To quantitatively summarize hotspot concentration across all eight genes, the proportion of the 263 mutated residue positions meeting the recurrence threshold was calculated for each gene and displayed as a bar plot (Figure 5).

Figure 5
Recurrent Residue Fraction per Gene (N = 263 Unique Residue Positions)



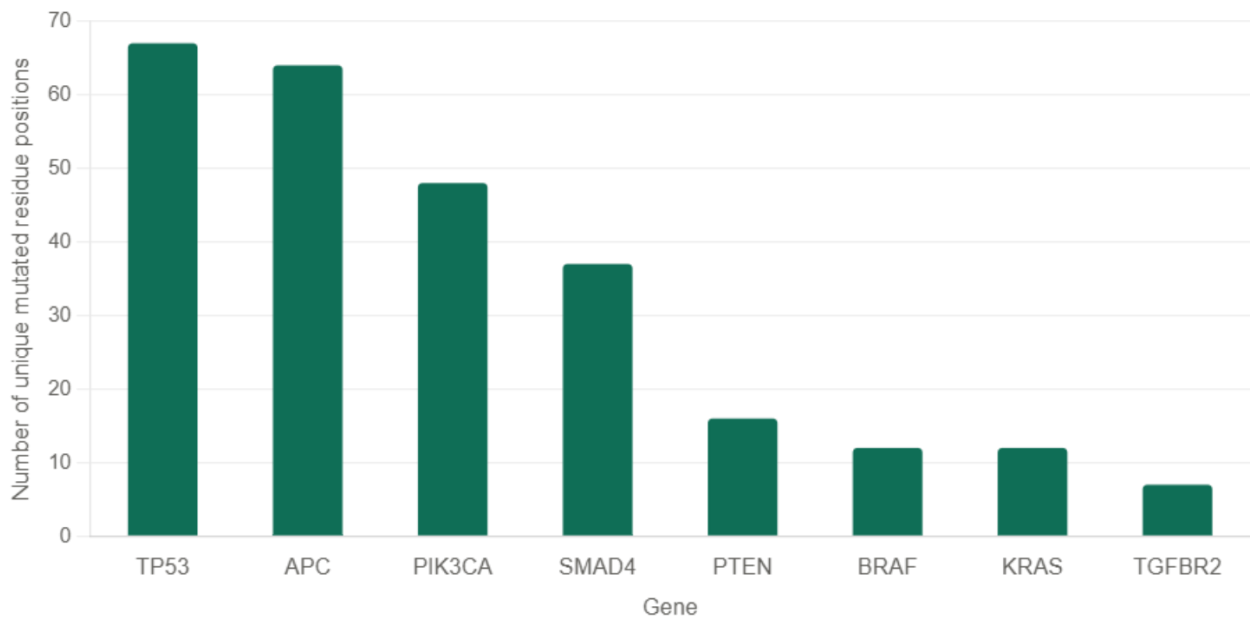
Note. Genes ordered by descending recurrent fraction.

KRAS exhibited the highest recurrent residue fraction, with approximately one quarter of its mutated positions classified as recurrent. In contrast, APC and TGFBR2 exhibited recurrent fractions of zero, with no individual residue position accumulating sufficient recurrences to meet the threshold despite APC exhibiting the highest overall mutation burden in the cohort (Table 2). These results reinforce the architectural distinctions observed in Figure 3 and demonstrate that a high gene-level mutation burden does not necessarily correspond to concentrated residue-level hotspot activity.

Mutated Residues per Gene

To assess mutational dispersion across protein sequences, Figure 6 presents the number of unique mutated residue positions identified per gene across the 263 residue-level observations. This metric reflects the number of distinct residue positions affected by at least one missense mutation.

Figure 6
Number of Unique Mutated Residue Positions per Gene Across the 263 Residue-Level Dataset



Note. Genes ordered by descending count. A high count reflects a dispersed mutational architecture; a low count reflects concentration at few positions.

TP53 exhibited the highest number of unique mutated residues, followed by APC. In contrast, KRAS contained comparatively few mutated residue positions despite its high recurrence levels. PIK3CA displayed an intermediate pattern, with mutations distributed across multiple residue positions while retaining several highly recurrent hotspots. Together, these findings demonstrate substantial differences in mutation dispersion and recurrence architecture across the analyzed driver genes.

4. Discussion

4.1 Study Overview and Main Findings

Cancer arises through stochastic evolutionary processes in which random mutations may confer a selective advantage, promoting clonal expansion and tumor progression (Fontana et al., 2023). From an evolutionary perspective, functionally important protein residues tend to be constrained, as natural selection removes deleterious substitutions over long timescales (Zhao et al., 2019). Although mutations arise largely at random, their persistence and repeated selection within tumors can be explained by somatic selection acting on functionally important and evolutionarily conserved residues, leading to recurrent mutation hotspots through convergent somatic evolution (Brown et al., 2019; Chang et al., 2016; Greaves, 2015).

Based on this framework, the present study investigated whether recurrent mutations in established colorectal cancer driver genes preferentially occur at evolutionarily conserved residues. By integrating mutation recurrence with conservation scores derived from multi-species sequence alignments, the analysis aimed to determine whether somatic selection preferentially targets functionally constrained sites within proteins. Although recurrent residues (Table 7) exhibited slightly higher mean conservation scores compared with non-recurrent residues, the difference was small and not statistically significant according to the Mann-Whitney test. Nevertheless, a weak but statistically significant positive correlation between recurrence frequency and conservation was observed in the Spearman analysis, suggesting a modest tendency for more frequently mutated residues to occur at highly conserved positions.

4.2 Statistical Interpretation of the Association Between Mutation Recurrence and Conservation

The absence of a statistically significant difference in conservation scores between recurrent and non-recurrent residues can be attributed to several factors. First, both the Mann-Whitney comparison and the logistic regression model rely on a binary classification of residues as recurrent

or non-recurrent. This classification ignores the magnitude of recurrence and therefore reduces the information contained in the dataset. For example, residues with recurrence counts of 5 and 57 are treated identically once categorized as recurrent. This likely contributed to the lack of a strong statistical signal despite the consistently positive direction of the association observed across all three analyses. The logistic regression coefficient was positive (coef = 6.215), the mean conservation score was higher in recurrent residues (Table 7), and the Spearman correlation confirmed a weak but significant positive trend ($\rho = 0.139$, $p = 0.025$). The binary classification therefore appears to have limited the ability to detect an association that the Spearman analysis, which retained the full recurrence magnitude, was able to capture (Altman & Royston, 2006).

A second contributing factor is the limited size of the recurrent group. Of the 263 mutated residues analyzed, only 15 met the recurrence threshold, whereas 248 were classified as non-recurrent. Such imbalance reduces statistical power and limits the ability of distribution-based tests to detect meaningful biological differences between groups. Additionally, conservation scores showed limited variability across the dataset, with the majority of residues falling between 0.85 and 1.00. As a result, the conservation score distributions of the two groups appeared highly similar to the statistical tests, even though recurrent residues exhibited a slightly higher mean value (Table 7). This ceiling effect in conservation scores reflects the biological nature of the genes analyzed. The eight genes included in this study are well-established colorectal cancer driver genes with critical roles in tumorigenesis, including oncogenes such as KRAS and tumor suppressor genes such as APC and TP53. Because these genes perform fundamental cellular functions, their residues are subject to strong purifying selection across species, resulting in high conservation scores regardless of recurrence status (Zhao et al., 2019). These genes participate in essential cellular pathways controlling proliferation, genomic stability, and signaling. For instance, activating mutations in KRAS, which occur in approximately 40 to 50% of colorectal cancers, promote constitutive signaling through pathways regulating cell proliferation (Yildirim et al., 2025). Similarly, loss-of-function mutations in APC disrupt regulation of beta-catenin within the Wnt signaling pathway, while TP53 mutations impair genomic surveillance mechanisms (Munteanu & Mastalier, 2014). Taken together, the observed results remain consistent with the broader literature describing the critical functional roles of these genes in colorectal cancer.

4.3 Distinct Mutation Architectures of Oncogenes and Tumor Suppressor Genes

Oncogenes typically contribute to tumorigenesis through gain-of-function alterations that increase or modify the activity of the encoded protein (Munteanu & Mastalier, 2014). In most cases, only a limited number of specific missense mutations can generate this activating effect by disrupting protein-protein interactions, altering allosteric regulation, affecting post-translational modification sites, or modifying ligand binding. As a result, activating mutations in oncogenes tend to cluster at specific functional residues, producing the well-known hotspot mutation architecture (Baeissa et al., 2016; Chillón-Pino et al., 2024). This pattern was clearly observed in the three oncogenes analyzed in this study. The most recurrent mutation identified in this cohort was KRAS G12D, observed in 57 independent tumors (Table 3). Additional substitutions at the same codon, including G12V (49 recurrences) and G13D (37 recurrences), were also among the most recurrent in the dataset (not shown). The concentration of mutations at positions 12 and 13 is visible in Figure 4 as the two dominant peaks in the KRAS scatter plot, and in Figure 3 where KRAS shows two residue positions in the highest recurrence category, consistent with the literature identifying these codons as the primary mutational hotspots in KRAS in colorectal cancer (Pellatt et al., 2025). These mutations disrupt the balance between the inactive GDP-bound and active GTP-bound forms of the KRAS protein, resulting in constitutive activation of downstream signaling pathways that promote cell proliferation and survival (Navarro-Jiménez et al., 2025).

A similar hotspot pattern was observed for BRAF, another proto-oncogene involved in the MAPK signaling pathway. The BRAF V600E mutation was the most recurrent alteration in this gene in the present cohort, observed in 48 tumors (Table 3). This mutation is known to produce a constitutively active kinase that drives continuous signaling through the RAS/RAF/MEK/ERK cascade and accounts for approximately 90% of BRAF alterations in metastatic colorectal cancer (Guerrero et al., 2022). Likewise, PIK3CA exhibited recurrent mutations at known activating residues, including E545K, the most recurrent substitution in this gene with 35 recurrences (Table 3), and H1047R (not shown), both of which alter the activity of the PI3K lipid kinase responsible for generating PIP3 and activating downstream growth-promoting pathways including AKT/mTOR signaling (Ranjbar et al., 2019; Wang et al., 2024). Although the mutation distribution in PIK3CA appeared somewhat more dispersed than in KRAS or BRAF, as reflected

by the broader spread of dots across residue positions in Figure 4 and the higher number of single-occurrence residues in Figure 3, several recurrent hotspots were still evident. Together, these results are consistent with the expected mutation architecture of oncogenes, where a small number of specific activating substitutions confer strong selective advantages and therefore recur across independent tumors.

In contrast, tumor suppressor genes contribute to tumorigenesis primarily through loss of function (Baeissa et al., 2016). Because protein inactivation can result from many different types of mutation, including nonsense, frameshift, and missense alterations, anywhere along the protein sequence, mutations tend to be dispersed rather than concentrated at specific hotspots (Baeissa et al., 2016; Chillón-Pino et al., 2024). This is reflected in the data: APC showed the highest overall mutation burden at the gene level (Table 2) and the highest total mutation events across all subtypes (Table 4), yet when the analysis was restricted to missense mutations only, APC contributed just 67 events (Table 6), the lowest among the more frequently mutated genes. Furthermore, Figure 6 shows that those 67 missense events were distributed across 64 unique residue positions, confirming the dispersed architecture. APC mutations frequently produce truncated proteins that disrupt regulation of the Wnt/beta-catenin signaling pathway, representing one of the earliest events in colorectal tumorigenesis (Wang et al., 2024). This pattern is consistent with findings reported by Baeissa et al. (2016) and Li et al. (2024), which demonstrate that tumor suppressor inactivation can result from a wide variety of truncating mutations distributed across the gene, unlike oncogenes which require specific activating substitutions at defined residues.

The tumor suppressor TP53 displayed a distinct architecture that partially resembles that of oncogenes. Unlike most tumor suppressor genes, TP53 mutations in colorectal cancer are predominantly missense rather than truncating (Li et al., 2024), which is reflected in Table 6 where TP53 contributed the highest number of missense events of all eight genes (235 events, 27.8%). These missense alterations often occur within the DNA-binding domain and result in structural changes that impair the ability of p53 to regulate transcriptional responses involved in DNA repair, apoptosis, and cell cycle arrest (Zsolt Kovács et al., 2024). TP53 mutations occur in approximately 45 to 54% of colorectal tumors (Nejad & Yaghoobi, 2025; Zsolt Kovács et al., 2024), which explains why TP53 also exhibited the highest number of unique mutated residue positions across

the dataset (Figure 6). At the same time, several well-known hotspot mutations were identified, including R175H, the most recurrent TP53 substitution in this cohort with 32 recurrences (Table 3), and R248Q, R273H, and R282W (not shown), consistent with established hotspots reported in the literature (Li et al., 2024) and visible in Figure 4. The difference between Table 6 and Figure 6 for TP53 is therefore biologically meaningful: the high missense event count reflects the frequency with which TP53 is mutated across the cohort, while the high unique position count reflects the fact that those mutations are spread across many different residues rather than concentrated, the hallmark of a gene where many different changes can disrupt function. This combination of dispersed mutations alongside recurrent hotspots reflects the unique architecture of TP53, which can exhibit both classical loss-of-function tumor suppressor behavior and, in some cases, gain-of-function properties associated with mutant p53 (Hassin et al., 2022; Li et al., 2024).

Taken together, the mutation architectures observed across these genes provide evidence of positive selection during tumor evolution. Although mutations arise largely at random, those that confer a proliferative or survival advantage are preferentially retained and expanded within tumor cell populations (Greaves, 2015; Navarro-Jiménez et al., 2025). The repeated occurrence of identical mutations such as KRAS G12D and BRAF V600E across independent tumors represents convergent somatic evolution, in which separate tumors independently acquire the same advantageous alterations. Despite the underlying genetic heterogeneity of tumors, these selective processes drive cancer cells toward similar functional outcomes through different genetic routes (Dasari et al., 2021; Yates & Campbell, 2012).

4.4 Evolutionary Conservation as an Incomplete Predictor of Mutation Recurrence

Although evolutionary conservation can serve as a proxy for functional importance (Murphy et al., 2022), conservation alone was not sufficient to fully explain mutation recurrence in this cohort. Many residues across these proteins are highly conserved yet only a subset become recurrent mutation sites, suggesting that additional factors including protein structural context, signaling pathway involvement, and underlying mutational mechanisms also influence whether a mutation confers a selective advantage during tumor evolution (Brown et al., 2019).

4.5 Broader Evolutionary Context

If evolutionary conservation alone does not fully explain mutation recurrence, additional biological layers must be considered. Cancer evolution operates through context-dependent selection, where the fitness advantage of a mutation depends not on its intrinsic properties alone but on the genetic and environmental background of the tumor (Greaves, 2015; McFarland et al., 2017). Epistatic interactions illustrate this principle, as driver mutations do not act in isolation and their selective impact may depend on mutations already present in the cell. Even when recurrent mutations affect evolutionarily conserved residues, their functional consequences may still vary depending on the tumor microenvironment and the signaling pathways involved (Brown et al., 2019; Greaves, 2015). These considerations help explain why conservation captures only part of the complexity underlying mutation recurrence.

A further contributing factor is the convergence of cancer cell states. Even when tumors acquire different driver mutations, they often converge toward similar cellular programs and gene expression profiles that enable tumor growth and progression. Studies across multiple cancer types including colorectal cancer have shown that tumors with distinct driver mutations frequently display similar transcriptional states along the developmental hierarchy of the cell of origin (Patel & Yanai, 2024). This suggests that cancer evolution repeatedly arrives at a limited set of functional solutions required to sustain tumor growth, reinforcing the idea that recurrence patterns cannot be fully explained by evolutionary conservation alone.

5. Translational Implications and Implementation Considerations

5.1 Diagnostic Feasibility Based on Mutation Architecture

The distinct mutational architectures observed across the eight driver genes in this study have direct relevance to how these genes are approached in molecular diagnostic settings. Genes in which mutations cluster at a small number of amino acid positions, such as KRAS and BRAF, where the vast majority of recurrent mutations concentrate at specific codons, are well suited to focused, codon-specific detection methods such as real-time PCR. In these cases, a single assay can be designed to interrogate only the known hotspot positions. In contrast, genes exhibiting dispersed mutation patterns with no strongly recurrent residues, such as APC and TGFBR2, cannot be efficiently captured by targeted assays that interrogate only specific positions. Their detection requires broader sequencing approaches capable of examining the entire coding sequence. Although the findings of this study are derived from the TCGA dataset, similar gene-level mutation frequency patterns have been reported in other colorectal cancer cohorts, including Arab patient populations analysed using next-generation sequencing, suggesting that these architectural differences are consistent across populations rather than being unique to this dataset (Al-Shamsi et al., 2016).

5.2 The Commercial Diagnostic Landscape

Molecular profiling in colorectal cancer is now a clinical standard rather than a research tool. The NCCN Clinical Practice Guidelines (v.3.2024) recommend that all patients with metastatic colorectal cancer undergo genotyping for KRAS, NRAS, and BRAF mutations before initiating systemic therapy (NCCN, 2024). Patients with KRAS mutations generally do not benefit from anti-EGFR therapies such as cetuximab or panitumumab, whereas patients with BRAF V600E mutations may be eligible for targeted therapeutic combinations involving BRAF inhibitors (Guerrero et al., 2022; NCCN, 2024). In the present study, KRAS and BRAF were not only the genes displaying the most concentrated hotspot architectures but also the genes whose mutations most directly influence treatment decisions. This clinical relevance is one reason why they are prioritised in companion diagnostic development. The companies operating in this space offer

broad precision oncology portfolios, of which colorectal cancer testing represents only one component. Three companies are particularly relevant to the findings of this study: Foundation Medicine, Illumina, and QIAGEN.

Foundation Medicine offers two products that illustrate the increasing integration of genomic profiling into clinical oncology. FoundationOne CDx is a comprehensive NGS assay that analyses 324 cancer-related genes, microsatellite instability status, and tumour mutational burden. Its liquid biopsy counterpart, FoundationOne Liquid CDx, analyses circulating tumour DNA extracted from blood rather than tumour tissue and received FDA approval in January 2026 for detecting BRAF V600E mutations to guide treatment with encorafenib plus cetuximab (Targeted Oncology, 2026; Yaeger et al., 2025).

Illumina's TruSight Oncology Comprehensive, approved by the FDA in August 2024, profiles 517 genes across solid malignant tumors and carries companion diagnostic indications for NTRK fusion positive solid tumors and RET fusion positive non-small cell lung cancer. Although the assay covers clinically relevant colorectal cancer genes, its colorectal cancer-specific companion diagnostic indications are still being established (Illumina, 2024; FDA, 2024a).

In contrast to these broad NGS platforms, QIAGEN's therascreen KRAS RGQ PCR Kit (QIAGEN, 2025) adopts a focused, codon-specific strategy. This FDA-approved real-time PCR assay detects seven specific mutations at KRAS codons 12 and 13 from formalin-fixed paraffin-embedded (FFPE) tumor tissue. Although substantially narrower in scope than comprehensive NGS panels, it remains commercially valuable because KRAS mutations are both clinically important and highly concentrated within a small number of hotspots. This observation is directly consistent with the findings of the present study: KRAS G12D was observed in 57 independent tumours, G12V in 49, and G13D in 37. Because most clinically relevant KRAS mutations occur at only a handful of codons, sequencing hundreds of genes is not always necessary to determine whether a patient carries a treatment-relevant KRAS mutation. Furthermore, as new KRAS-targeted therapies have gained regulatory approval, including sotorasib plus panitumumab and adagrasib plus cetuximab for KRAS G12C-mutant colorectal cancer, the same kit has accumulated multiple companion diagnostic indications (FDA, 2024b; QIAGEN, 2025). This demonstrates that a highly

concentrated hotspot architecture can support a commercially successful and clinically impactful diagnostic strategy without requiring comprehensive genomic profiling.

Taken together, these products illustrate that colorectal cancer diagnostics span a broad spectrum, ranging from rapid hotspot-focused assays to large-scale comprehensive genomic profiling platforms. Each occupies a distinct clinical and operational niche that reflects, at least in part, the underlying mutational architecture of the genes being interrogated.

5.3 Gene Prioritisation and a Framework for Panel Design

To translate the mutation architecture findings of this study into a structured prioritisation framework, a decision matrix approach was applied to rank the eight driver genes according to their suitability for inclusion in a targeted colorectal cancer diagnostic panel. This method ensures that the ranking is driven by the data rather than by prior assumptions regarding which genes are most important (Seshaiyer, 2022). Four criteria were used to evaluate each gene: hotspot concentration (defined as the proportion of mutated residues meeting the ≥ 5 tumour recurrence threshold), maximum recurrence (the highest recurrence value observed for a single mutation, used as a proxy for signal strength), conservation of recurrent residues (reflecting evolutionary constraint at frequently mutated positions), and clinical actionability (based on whether mutations in the gene influence eligibility for approved targeted therapies according to NCCN guidelines). For the clinical actionability criterion, a score of 3 was assigned to genes with approved companion diagnostics, 2 to genes with prognostic significance or emerging biomarker status, and 1 to genes with no currently approved targeted therapy (NCCN, 2024). The resulting prioritisation framework is presented in Table 8.

Table 8*Decision Matrix for Gene Prioritisation in a Targeted CRC Diagnostic Panel*

Gene	Hotspot concentration (1–3)	Max recurrence (1–3)	Conservation of recurrent residues (1–3)	Clinical actionability (1–3)	Total (max 12)
KRAS	3	3	3	3	12
BRAF	2	3	3	3	11
PIK3CA	2	2	3	2	9
TP53	2	2	2	1	7
SMAD4	1	1	2	1	5
PTEN	1	1	2	1	5
APC	1	2	1	1	5
TGFBR2	1	1	1	1	4

Note. Scoring: 3 = high / most favourable; 2 = moderate; 1 = low / least favourable. Hotspot concentration scored on recurrent residue fraction (KRAS 0.25, BRAF/PIK3CA 0.083, TP53 0.074, all others ≤ 0.062 or 0). Max recurrence: 3 = ≥ 40 tumours; 2 = 10–39; 1 = <10 . Clinical actionability: 3 = directly determines eligibility for an approved targeted therapy (NCCN, 2024); 2 = prognostic or emerging biomarker; 1 = no approved targeted therapy currently available. Source: author’s own analysis.

The decision matrix identifies KRAS and BRAF as the highest-priority genes, achieving the strongest overall scores across all four criteria. Both genes exhibit concentrated hotspot architectures, contain highly recurrent mutations, and directly influence treatment decisions through approved targeted therapies. These characteristics make them particularly suitable candidates for focused diagnostic assays and help explain why they have become central targets in companion diagnostic development. PIK3CA ranked third, reflecting moderate hotspot concentration and strong conservation at recurrent sites. TP53 and SMAD4 occupy an intermediate

tier; their mutations carry prognostic significance, but dispersed mutation profiles and the absence of currently approved targeted therapies reduce their priority for inclusion in a focused initial panel. APC and TGFBR2 scored lowest, with no residues meeting the recurrence threshold, no approved targeted agents, and, in APC's case, the additional practical challenge of a protein spanning 2,843 amino acids, which renders any hotspot-based approach unsuitable regardless of its biological importance.

It is important to note that this ranking reflects diagnostic tractability rather than biological significance, and the two are not synonymous. APC is one of the most frequently mutated genes in the entire dataset and plays a fundamental role in colorectal tumorigenesis. Its low ranking within this framework indicates only that its dispersed mutation architecture makes it poorly suited to targeted hotspot detection, not that it is unimportant as a driver gene.

5.4 Economic and Logistical Considerations

Taken together, the findings of this study suggest a clear economic and logistical rationale for matching diagnostic strategy to mutation architecture. According to a systematic review of RAS genetic testing in metastatic colorectal cancer, a combined KRAS and BRAF PCR-based test costs approximately Int\$323 per patient, compared with approximately Int\$4,016 for comprehensive NGS profiling, representing a twelvefold difference (Unim et al., 2020). For genes such as KRAS, whose clinically relevant variants are concentrated at codons 12 and 13, a targeted PCR assay can detect the mutations that directly influence treatment decisions at a fraction of the cost of sequencing hundreds of genes. Conversely, for genes such as APC and TGFBR2, which exhibit no recurrent hotspots, a targeted PCR assay would miss the majority of clinically relevant variants. In these cases, broader sequencing coverage is not a luxury but a necessity imposed by the underlying mutation architecture (Unim et al., 2020).

Based on the mutation architecture data presented in this study, a reasonable diagnostic approach for colorectal cancer patients requiring molecular profiling for therapy selection would be to prioritise a focused PCR-based test for KRAS and BRAF as an initial step, covering the mutations already recommended by NCCN guidelines and detected by commercially approved companion

diagnostic assays. Broader NGS panel testing could then be reserved for situations requiring comprehensive tumour characterisation, assessment of clinical trial eligibility across multiple genes, or evaluation of tumour suppressor genes with dispersed mutation profiles. This is not a novel clinical pathway but rather an evidence-based rationale for the tiered testing approach already used in clinical practice, now supported by the residue-level mutation architecture and evolutionary conservation data presented in this study.

A further logistical consideration is reimbursement. FDA approval and payer coverage are separate processes, and the gap between them can have direct consequences for patient access. Assays included within reimbursement policies will generally reach more patients than newly approved alternatives of equivalent or even superior analytical performance because uninsured testing costs create barriers that many patients and healthcare systems cannot absorb. The UnitedHealthcare Community Plan medical policy effective March 2026 illustrates this fragmentation concretely: FoundationOne CDx is covered for colorectal cancer for cetuximab and panitumumab indications, FoundationOne Liquid CDx is covered for BRAF V600E-mutant metastatic colorectal cancer to guide encorafenib plus cetuximab, and MI Cancer Seek holds coverage for both BRAF V600E and KRAS wild-type indications in colorectal cancer, yet Guardant360 CDx does not yet have a colorectal cancer-specific covered indication in this policy despite receiving FDA approval in January 2026 (UHC, 2026). From an implementation perspective, reimbursement status is often a more decisive determinant of clinical uptake than analytical validity alone (Schramm et al., 2024)

5.5 Future Directions

The most practical contribution of this study is not the development of a new standalone assay but rather the provision of an additional layer of evidence to inform gene prioritisation within existing colorectal cancer testing panels and to guide whether a hotspot-focused or sequencing-based approach is most appropriate for a given gene. As ctDNA-based platforms continue to mature, the same mutation architecture principles may extend to non-invasive monitoring applications, with hotspot genes such as KRAS remaining particularly tractable targets (Zeng et al., 2024; Marsoni et al., 2025). As targeted therapy options expand and the regulatory and reimbursement landscape becomes increasingly complex, systematic and evidence-based approaches for prioritising which

mutations to detect, and how to detect them, will become increasingly important. The mutation architecture analysis presented here represents one contribution towards that objective

References

- Al-Shamsi, H. O., Jones, J., Fahmawi, Y., Dahbour, I., Tabash, A., Abujamous, L., Lau, M. W., Hassan, M., Ali, J., Husain, H., & Kurzrock, R. (2016). Molecular spectrum of KRAS, NRAS, BRAF, PIK3CA, TP53, and APC somatic gene mutations in Arab patients with colorectal cancer: Determination of frequency and distribution pattern. *Journal of Gastrointestinal Oncology*, 7(6), 875–883. <https://doi.org/10.21037/jgo.2016.08.08>
- Altman, D. G., & Royston, P. (2006). The cost of dichotomising continuous variables. *BMJ*, 332(7549), 1080. <https://doi.org/10.1136/bmj.332.7549.1080>
- Baeissa, H. M., Benstead-Hume, G., Richardson, C. J., & Pearl, F. M. G. (2016). Mutational patterns in oncogenes and tumour suppressors. *Biochemical Society Transactions*, 44(3), 925–931. <https://doi.org/10.1042/bst20160001>
- Bányai, L., Trexler, M., Kerekes, K., Csuka, O., & Patthy, L. (2021). Use of signals of positive and negative selection to distinguish cancer genes and passenger genes. *ELife*, 10. <https://doi.org/10.7554/elife.59629>
- Brown, A.-L., Li, M., Goncarenco, A., & Panchenko, A. R. (2019). Finding driver mutations in cancer: Elucidating the role of background mutational processes. *PLOS Computational Biology*, 15(4), e1006981. <https://doi.org/10.1371/journal.pcbi.1006981>
- Cerami, E., Gao, J., Dogrusoz, U., Gross, B. E., Sumer, S. O., Aksoy, B. A., Jacobsen, A., Byrne, C. J., Heuer, M. L., Larsson, E., Antipin, Y., Reva, B., Goldberg, A. P., Sander, C., & Schultz, N. (2012). The cBio Cancer Genomics Portal: An open platform for exploring multidimensional cancer genomics data. *Cancer Discovery*, 2(5), 401–404. <https://doi.org/10.1158/2159-8290.CD-12-0095>
- Chang, M. T., Asthana, S., Gao, S. P., Lee, B. H., Chapman, J. S., Kandoth, C., Gao, J., Socci, N. D., Solit, D. B., Olshen, A. B., Schultz, N., & Taylor, B. S. (2016). Identifying recurrent mutations in cancer reveals widespread lineage diversity and mutational specificity. *Nature Biotechnology*, 34(2), 155–163. <https://doi.org/10.1038/nbt.3391>
- Chen, Y., Feng, X., Reid, K., Zhang, C., Löytynoja, A., & Merilä, J. (2025). Dynamics of deleterious mutations and purifying selection in small population isolates. *Molecular Biology and Evolution*, 42(7), msaf110. <https://doi.org/10.1093/molbev/msaf110>

- Chillón-Pino, D., Badonyi, M., Semple, C. A., & Marsh, J. A. (2024). Protein structural context of cancer mutations reveals molecular mechanisms and candidate driver genes. *Cell Reports*, 43(11), 114905. <https://doi.org/10.1016/j.celrep.2024.114905>
- Dasari, K., Somarelli, J. A., Kumar, S., & Townsend, J. P. (2021). The somatic molecular evolution of cancer: Mutation, selection, and epistasis. *Progress in Biophysics and Molecular Biology*, 165, 56–65. <https://doi.org/10.1016/j.pbiomolbio.2021.08.003>
- Fontana, D., Crespiatico, I., Crippa, V., Malighetti, F., Villa, M., Angaroni, F., De Sano, L., Aroldi, A., Antoniotti, M., Caravagna, G., Piazza, R., Graudenzi, A., Mologni, L., & Ramazzotti, D. (2023). Evolutionary signatures of human cancers revealed via genomic analysis of over 35,000 patients. *Nature Communications*, 14(1), 5982. <https://doi.org/10.1038/s41467-023-41670-3>
- Gao, J., Aksoy, B. A., Dogrusoz, U., Dresdner, G., Gross, B., Sumer, S. O., Sun, Y., Jacobsen, A., Sinha, R., Larsson, E., Cerami, E., Sander, C., & Schultz, N. (2013). Integrative analysis of complex cancer genomics and clinical profiles using the cBioPortal. *Science Signaling*, 6(269), pl1. <https://doi.org/10.1126/scisignal.2004088>
- Greaves, M. (2015). Evolutionary determinants of cancer. *Cancer Discovery*, 5(8), 806–820. <https://doi.org/10.1158/2159-8290.cd-15-0439>
- Guardant Health. (2026, January 22). *Guardant Health receives FDA approval for Guardant360 CDx as companion diagnostic for BRAFTOVI (encorafenib) combination in patients with BRAF V600E-mutant metastatic colorectal cancer* [Press release]. <https://investors.guardanthealth.com/press-releases/press-releases/2026/Guardant-Health-Receives-FDA-Approval-for-Guardant360-CDx-as-Companion-Diagnostic-for-BRAFTOVI-encorafenib-Combination-in-Patients-with-BRAF-V600E-Mutant-Metastatic-Colorectal-Cancer/default.aspx>
- Guerrero, R. M., Arnao Labajos, V., Lozano Ballena, S., Aliaga Macha, C., Sotelo Lezama, M., Pacheco Roman, C., Montenegro Beltran, P., & Figueroa Torrejon, A. (2022). Targeting BRAF V600E in metastatic colorectal cancer: Where are we today? *Ecancermedicalscience*, 16, 1489. <https://doi.org/10.3332/ecancer.2022.1489>

- Hassin, O., Nataraj, N. B., Shreberk-Shaked, M., Aylon, Y., Yaeger, R., Fontemaggi, G., Mukherjee, S., Maddalena, M., Avioz, A., Iancu, O., Mallel, G., Gershoni, A., Grosheva, I., Feldmesser, E., Ben-Dor, S., Golani, O., Hendel, A., Blandino, G., Kelsen, D., & Yarden, Y. (2022). Different hotspot p53 mutants exert distinct phenotypes and predict outcome of colorectal cancer patients. *Nature Communications*, 13(1), 2800. <https://doi.org/10.1038/s41467-022-30481-7>
- Hoyos, D., Zappasodi, R., Schulze, I., Sethna, Z., de Andrade, K. C., Bajorin, D. F., Bandlamudi, C., Callahan, M. K., Funt, S. A., Hadrup, S. R., Holm, J. S., Rosenberg, J. E., Shah, S. P., Vázquez-García, I., Weigelt, B., Wu, M., Zamarin, D., Campitelli, L. F., Osborne, E. J., & Klinger, M. (2022). Fundamental immune–oncogenicity trade-offs define driver mutation fitness. *Nature*, 606(7912), 172–179. <https://doi.org/10.1038/s41586-022-04696-z>
- Illumina. (2024). *TruSight Oncology Comprehensive*. <https://www.illumina.com/products/by-type/ivd-products/trusight-oncology-comprehensive-us.html>
- Klimeck, L., Heisser, T., Hoffmeister, M., & Brenner, H. (2023). Colorectal cancer: A health and economic problem. *Best Practice & Research Clinical Gastroenterology*, 66, 101839. <https://doi.org/10.1016/j.bpg.2023.101839>
- Li, W., Li, L., Yang, H., Shi, C., Lei, Z., Guo, L., & Wang, Y. (2024). Unraveling the role of TP53 in colorectal cancer therapy: From wild-type regulation to mutant. *Frontiers in Bioscience-Landmark*, 29(7), 272. <https://doi.org/10.31083/j.fbl2907272>
- Marsoni, S., Siena, S., Bardelli, A., & Patelli, G. (2025). Clinical utility and future perspectives of liquid biopsy in colorectal cancer. *Communications Medicine*, 5(1), 137. <https://doi.org/10.1038/s43856-025-00852-4>
- McFarland, C. D., Yaglom, J. A., Wojtkowiak, J. W., Scott, J. G., Morse, D. L., Sherman, M. Y., & Mirny, L. A. (2017). The damaging effect of passenger mutations on cancer progression. *Cancer Research*, 77(18), 4763–4772. <https://doi.org/10.1158/0008-5472.can-15-3283-t>
- Munteanu, I., & Mastalier, B. (2014). Genetics of colorectal cancer. *Journal of Medicine and Life*, 7(4), 507. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4316127/>

- Murphy, D. A., Elyashiv, E., Amster, G., & Sella, G. (2022). Broad-scale variation in human genetic diversity levels is predicted by purifying selection on coding and non-coding elements. *ELife*, 12. <https://doi.org/10.7554/elife.76065>
- National Comprehensive Cancer Network. (2024). Colon cancer, version 3.2024, NCCN clinical practice guidelines in oncology. *Journal of the National Comprehensive Cancer Network*, 22(2D). <https://jnccn.org/view/journals/jnccn/22/2D/article-e240029.xml>
- Navarro-Jiménez, M., González, B., Mulet, N., Hierro, C., & Alonso, S. (2025). KRAS-targeted therapies in colorectal cancer: A systematic analysis of mutations, inhibitors, and clinical trials. *Npj Precision Oncology*, 9(1). <https://doi.org/10.1038/s41698-025-01166-3>
- Nejad, A. L., & Yaghoobi, M. M. (2025). Mutation analysis of TP53 tumor suppressor gene in colorectal cancer in patients from Iran (Kerman Province). *Iranian Journal of Basic Medical Sciences*, 15(1), 683. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3586867/>
- Nourbakhsh, M., Degn, K., Saksager, A., Tiberti, M., & Papaleo, E. (2024). Prediction of cancer driver genes and mutations: The potential of integrative computational frameworks. *Briefings in Bioinformatics*, 25(2). <https://doi.org/10.1093/bib/bbad519>
- Patel, A. S., & Yanai, I. (2024). A developmental constraint model of cancer cell states and tumor heterogeneity. *Cell*, 187(12), 2907–2918. <https://doi.org/10.1016/j.cell.2024.04.032>
- Pellatt, A. J., Bhamidipati, D., & Subbiah, V. (2025). Ready, set, go: Setting off on the mission to target KRAS in colorectal cancer. *OncoDaily*. <https://oncoday.com/oncolibrary/kRAS-g12d-and-g12v-mutations-in-crc>
- Persi, E., Sudalagunta, P. R., Wolf, Y. I., Canevarolo, R. R., Damaghi, M., Shain, K. H., Silva, A. S., & Koonin, E. V. (2025). Genome-level selection in tumors as a universal marker of resistance to therapy. *Nature Communications*, 16(1). <https://doi.org/10.1038/s41467-025-61709-x>
- QIAGEN. (2025). *therascreen KRAS RGQ PCR Kit*. <https://www.qiagen.com/us/products/diagnostics-and-clinical-research/oncology/therascreen-solid-tumor/therascreen-kRAS-rgq-pcr-kit-us>
- Ranjbar, R., Mohammadpour, S., Esfahani, A. T., Namazian, S., Yaghob-Taleghani, M., Baghaei, K., Mortazavi, A., Pasharavesh, L., & Nazemalhosseini-Mojarad, E. (2019). Prevalence

- and prognostic role of PIK3CA E545K mutation in Iranian colorectal cancer patients. *Gastroenterology and Hepatology from Bed to Bench*, 12(Suppl1), S22. <https://pmc.ncbi.nlm.nih.gov/articles/PMC7011074/>
- Schramm, W., Hollenbenders, Y., & Kurscheidt, M. (2024). Explorative cost-effectiveness analysis of colorectal cancer recurrence detection with next-generation sequencing liquid biopsy in Spain, France, and Germany. *Therapeutic Advances in Gastroenterology*, 17. <https://doi.org/10.1177/17562848241248246>
- Seshaiyer, P. (2022, March 1). Preparing future generations to address global pandemics with innovative mathematical thinking. *SIAM News*. <https://www.siam.org/publications/siam-news/articles/preparing-future-generations-to-address-global-pandemics-with-innovative-mathematical-thinking>
- Supek, F., Miñana, B., Valcárcel, J., Gabaldón, T., & Lehner, B. (2014). Synonymous mutations frequently act as driver mutations in human cancers. *Cell*, 156(6), 1324–1335. <https://doi.org/10.1016/j.cell.2014.01.051>
- Tang, Y.-Y., Wei, P.-J., Zhao, J., Xia, J., Cao, R.-F., & Zheng, C.-H. (2021). Identification of driver genes based on gene mutational effects and network centrality. *BMC Bioinformatics*, 22(S3). <https://doi.org/10.1186/s12859-021-04377-0>
- Targeted Oncology. (2026, January 15). FoundationOne Liquid CDx granted new FDA indication in mCRC. <https://www.targetedonc.com/view/foundationone-liquid-cdx-granted-new-fda-indication-in-mcrc>
- The Cancer Genome Atlas Research Network. (2018). Colorectal adenocarcinoma (TCGA, PanCancer Atlas) [Dataset]. cBioPortal. https://www.cbioportal.org/study/summary?id=coadread_tcg_pan_can_atlas_2018
- UnitedHealthcare Community Plan. (2026). *FDA companion diagnostic testing coverage policy (CS373.D)*. <https://www.uhcprovider.com/content/dam/provider/docs/public/policies/medicaid-comm-plan/fda-companion-diagnostic-testing-cs.pdf>
- Unim, B., Pitini, E., De Vito, C., D'Andrea, E., Marzuillo, C., & Villari, P. (2020). Cost-effectiveness of RAS genetic testing strategies in patients with metastatic colorectal cancer:

- A systematic review. *Value in Health*, 23(1), 114–126.
<https://doi.org/10.1016/j.jval.2019.07.009>
- U.S. Food and Drug Administration. (2024a). *TruSight Oncology Comprehensive (P230011)*.
<https://www.fda.gov/medical-devices/recently-approved-devices/trusight-oncology-comprehensive-p230011>
- U.S. Food and Drug Administration. (2024b). *PMA P110027: theascreen KRAS RGQ PCR Kit*.
https://www.accessdata.fda.gov/cdrh_docs/pdf11/P110027S012C.pdf
- Wang, H., Tang, R., Jiang, L., & Jia, Y. (2024). The role of PIK3CA gene mutations in colorectal cancer and the selection of treatment strategies. *Frontiers in Pharmacology*, 15.
<https://doi.org/10.3389/fphar.2024.1494802>
- Wollenberg Valero, K. C. (2020). Aligning functional network constraint to evolutionary outcomes. *BMC Evolutionary Biology*, 20(1). <https://doi.org/10.1186/s12862-020-01613-8>
- Wong, J. K. L., Aichmüller, C., Schulze, M., Hlevnjak, M., Elgaafary, S., Lichter, P., & Zapatka, M. (2022). Association of mutation signature effectuating processes with mutation hotspots in driver genes and non-coding regions. *Nature Communications*, 13(1), 178.
<https://doi.org/10.1038/s41467-021-27792-6>
- Yaeger, R., Martini, J.-F., Pasquina, L., Tunquist, B., Zhang, X., Kaiser, F., Pantoja Galicia, N., Deng, S., Gong, S., Guo, C., Kiely, J., Ng, T.-C. V., Ferrier, G., Tabernero, J., & Kopetz, S. (2025). Clinical validity of FoundationOne Liquid CDx for detection of BRAF V600E in colorectal cancer. *Cancer Research Communications*, 5(9), 1566–1573.
<https://doi.org/10.1158/2767-9764.CRC-25-0002>
- Yates, L. R., & Campbell, P. J. (2012). Evolution of the cancer genome. *Nature Reviews Genetics*, 13(11), 795–806. <https://doi.org/10.1038/nrg3317>
- Yildirim, H. C., Gunenc, D., Almuradova, E., Sutcuoglu, O., & Yalcin, S. (2025). A narrative review of RAS mutations in early-stage colorectal cancer: Mechanisms and clinical implications. *Medicina*, 61(3), 408. <https://doi.org/10.3390/medicina61030408>

- Zeng, Y., Mo, S., Gao, H., & Tan, X. (2024). Clinical application of liquid biopsy in colorectal cancer: Detection, prediction, and treatment monitoring. *Molecular Cancer*, 23, 145. <https://doi.org/10.1186/s12943-024-02063-2>
- Zhao, S., Liu, J., Nanga, P., Liu, Y., Cicek, A. E., Knoblauch, N., He, C., Stephens, M., & He, X. (2019). Detailed modeling of positive selection improves detection of cancer driver genes. *Nature Communications*, 10(1). <https://doi.org/10.1038/s41467-019-11284-9>
- Zsolt Kovács, Sugimura, H., György, T. A., Osvath, E., Manirakiza, F., & Gurzu, S. (2024). Bioinformatic identification of TP53 gene mutation hotspots in colorectal cancer. *International Journal of Molecular Sciences*, 25(12), 6612. <https://doi.org/10.3390/ijms25126612>